

Executive Summary:

Temporal Network Signals and Cross-Disciplinary Concept Spread

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Headline

Early co-occurrence neighbourhood openness (the OPEN index) predicts how many scientific fields a concept later appears in, beyond a five-feature popularity baseline. The non-selection evidence pool gives a partial Spearman correlation of +0.069 [+0.038, +0.100] ($I^2 = 0$, 6/6 bodies positive). The effect is small and exploratory: it adds no practical forecasting gain (+0.002 Spearman over baseline), the home-only variant is marginal (its CI includes zero at the strictest control rungs R4/R5 on the fresh cohort), and the run's within-concept closure test found no support for a causal mechanism.

Goal

Test whether early structural signals in a scientific concept's topic co-occurrence network anticipate its later cross-field breadth, measured by rarefied field breadth (O_{2r} , $m = 50$, hypergeometric rarefaction over venue fields in t_0+6 to t_0+8). The study screened 53 early network indicators across six families against a five-feature popularity baseline on held-out field groups, using a 12,499-concept panel drawn from the OpenAlex bulk snapshot (476M works, 2026-09-23).

What was tried (5 iterations, 25 artifacts, 21 completed)

Iteration 1 ran a wide screen of five rival spread mechanisms on a 48-concept dev panel: citation naturalisation (A_h^*), social reach, co-occurrence diversity (D_{ratio}), frequency-free selectivity, and gateway landing. None survived the pre-registered rule. Two artifacts failed (timeout). The gateway retention lead ($\Delta\text{AUC} +0.103$, 80 episodes) was the only forward signal.

Iteration 2 scaled to 12,499 concepts and 27,393 episodes with a hash-sealed held-out split. Gateway retention was decisively disconfirmed ($\Delta\text{AUC} = -0.00001$). A conditional-logit field-entry model confirmed that concepts spread to fields related to those retaining them ($d_0 = 0.30$, $p = 0.001$).

Iteration 3 delivered the 53-indicator held-out screen: 7 of 10 frozen indicators were confirmed for O_{2r} , spanning retained-frontier (4) and ego-network topology (3) families. The retained-frontier field-entry effect was PARTIAL ($d_0 = 0.322$ [0.291, 0.355] but volume-matched null and min-cp reversal). About half of the top two frontier indicators reflected pre-onset footprint.

Iteration 4 built the OPEN index (6 signed, z -scored ego-network components) and tested it on a never-screened 2015–2017 cohort: CONFIRMED but marginal (home-only PSP +0.091 at

R2, CI includes zero at R4/R5). The breadth decomposition showed 73% early exploration vs 27% retention. No discrete trajectory typology passed the naming rule (ARI = 0.222).

Iteration 5 tested the home-only churn signal on vocabulary-free Frame N concepts (frozen verdict PARTIAL: +0.117 [+0.020, +0.218] at R3), showed Cheng’s consistency is 98% current size, and decomposed the signal into partner classes: new-community partners from mixed-field papers carry it. The confound check (V2 permutation) revealed the signal is topical non-redundancy, not temporal turnover (6.5% retained).

Key results

1. **Seven held-out indicators confirmed** (3 ego-network, 4 frontier): the strongest are M0_density_end (+0.375), D_vol_end (+0.307), and CONTACT_REACH (+0.211). Half of the top two is pre-onset footprint.
2. **OPEN_{home} non-selection pool**: +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive (DL on Fisher z , HKSJ [+0.042, +0.096]).
3. **Retained-field relatedness predicts field entry**: $d_0 = 0.322$ [0.291, 0.355], 6,978 events, 3,162 concepts. Volume-matched contrast is null.
4. **Consistency reversal**: Cheng et al.’s +53.5%/SD drops to +1.3% after size control; as an early trait, consistency predicts narrower breadth (−0.069, 5/5 negative; selection data, not confirmed on Frame N, −0.064 with CI including zero).
5. **Breadth decomposition** (an accounting identity, unstratified variant): 77.9% early contact diversity, −4.6% frontier advance, 26.8% retention. Broad concepts start wide; they do not advance a wider frontier.
6. **Signal is static dispersion**: within-concept year-label permutation absorbs temporal churn (6.5% retained); the Chao and configuration-null composites retain 91–100%.

What failed and why

1. **Gateway retention** (iterations 1–2): dev-only artefact. Held-out $\Delta\text{AUC} = -0.00001$; the shuffled- R placebo’s 95th percentile exceeded the original.
2. **Citation naturalisation** A_h^* (iteration 1): $\Delta\rho = -0.006$; two-thirds of between-concept variance was general disciplinary homophily.
3. **Co-occurrence diversity** D_{ratio} (iteration 1): absorbed by the B5 baseline ($\rho_{B5} = 0.770$).
4. **Within-concept closure** $\rightarrow$ **entry** (iterations 4–5): null on DEV, opposite sign on held-out. No mechanism link established.
5. **Discrete trajectory typology** (iteration 4): DTW vs HMM ARI = 0.222. Breadth is a continuum.
6. **Four artifacts** timed out or failed validation across iterations 1, 3, and 4.

Open questions

1. **Causal mechanism**: association only. Early openness may reflect intrinsic generality, community diversity, or problem breadth rather than causing spread.

2. **Domain heterogeneity:** Life & Environment shows the weakest OPEN signal in the all-papers build (Physical Sciences is weakest in the home-only build); retained-field relatedness has $I^2 = 0.92$ across groups.
3. **Temporal turnover:** V2 excess has split-half reliability below 0.05, so the temporal-vs-static question is untested rather than answered.
4. **Vocabulary survivorship:** Frame N ($n = 636$, power 0.47) is small. Broader replication needed.
5. **Measurement noise:** OPEN_{home} split-half reliability is 0.49; individual prediction is imprecise.
6. **Practical utility:** the forecasting gain is +0.002 Spearman — negligible for any applied use.